

Ethanol Blending in Gasoline – Denmark

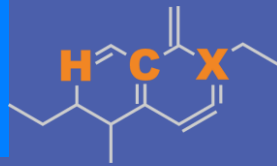
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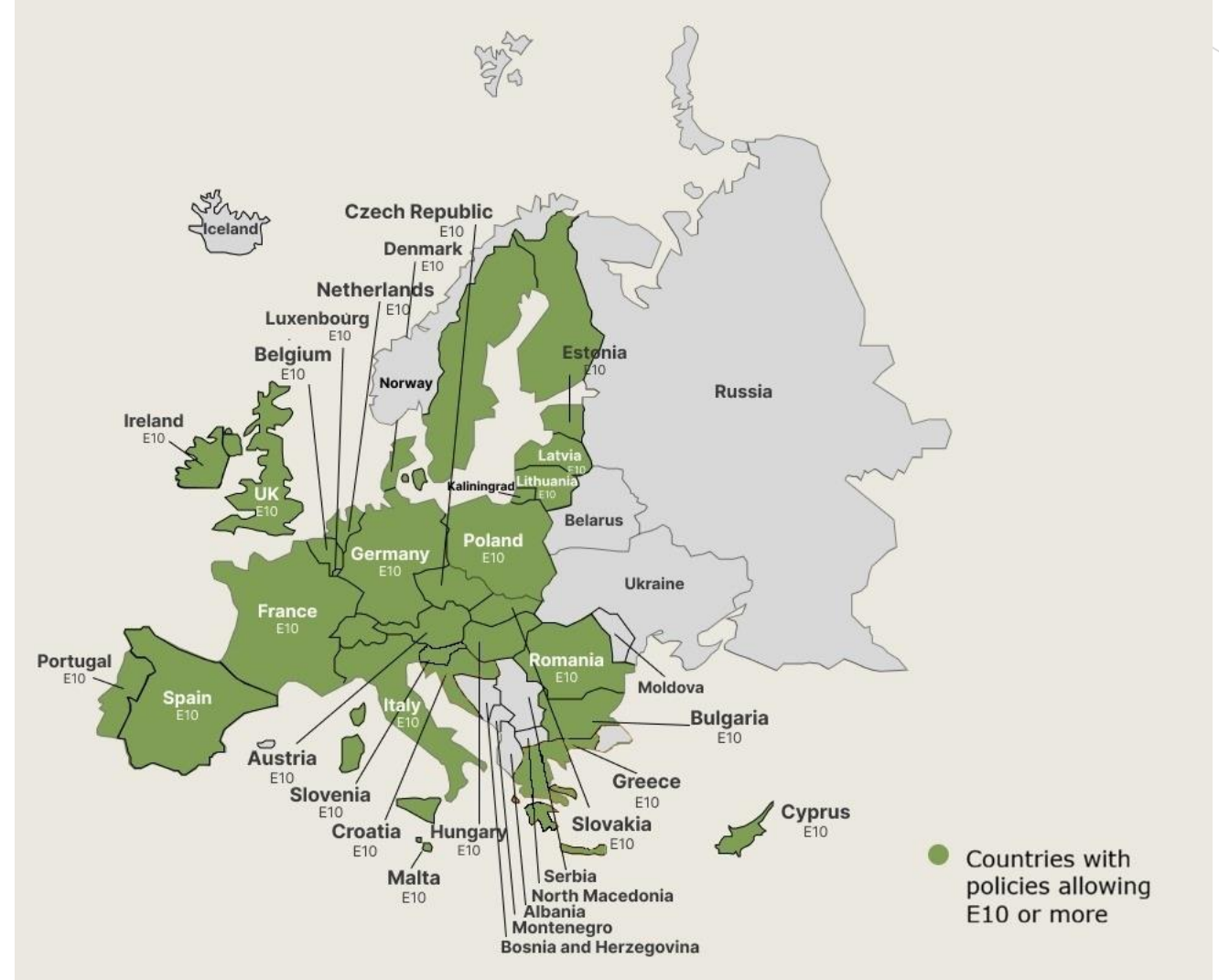


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Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Denmark

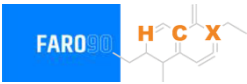


In 2024, gasoline consumption in Denmark was 1,415 million liters (374 million gallons) and growth rate was -2.6%. Gasoline production and net imports were 2,745 and -1,072 million liters (725 and -283 million gallons) correspondingly. Denmark complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.3 RON.

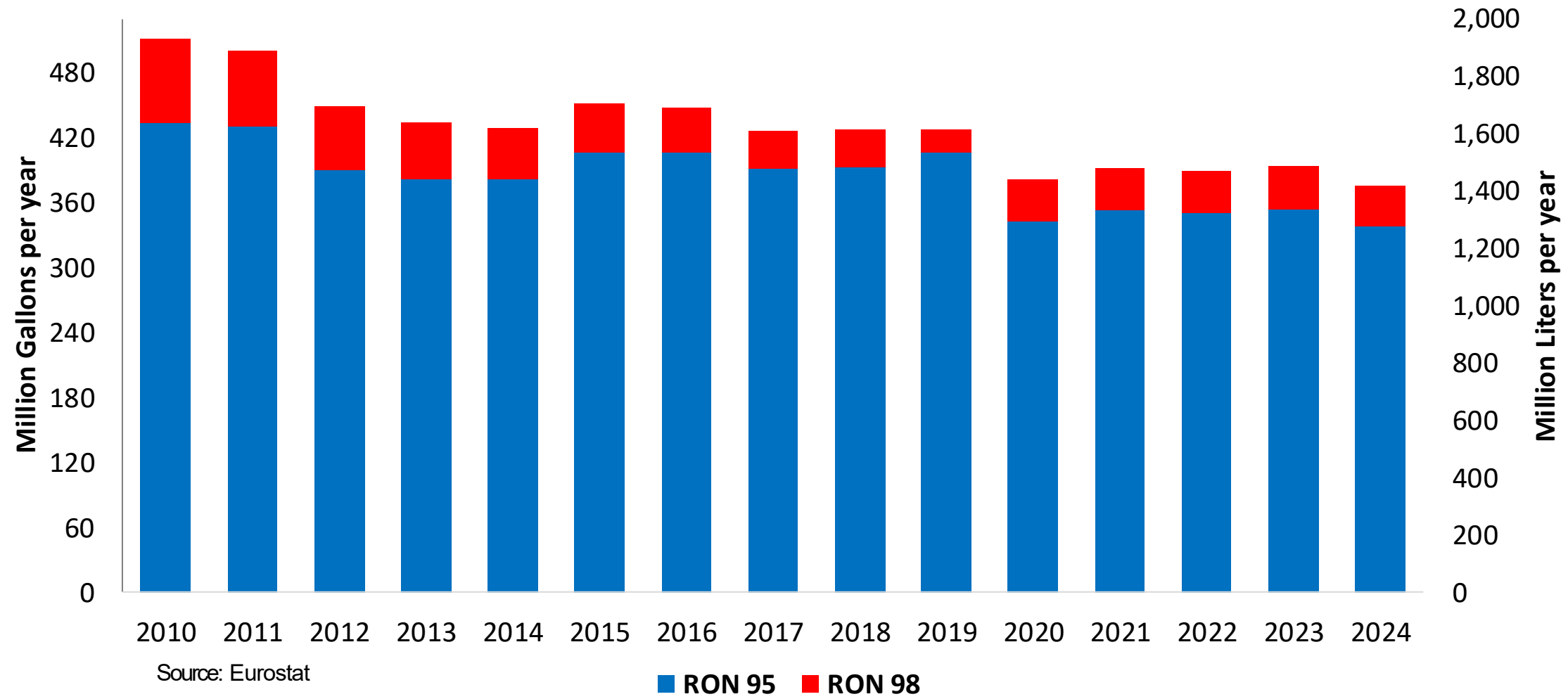
In 2024, ethanol consumption in Denmark was 168 million liters (44 million gallons) representing 11.9% of gasoline demand. Ethanol production and Net Imports were 160 and 8 million liters (44 and 2 million gallons) correspondingly.

Source: Eurostat, European Commission

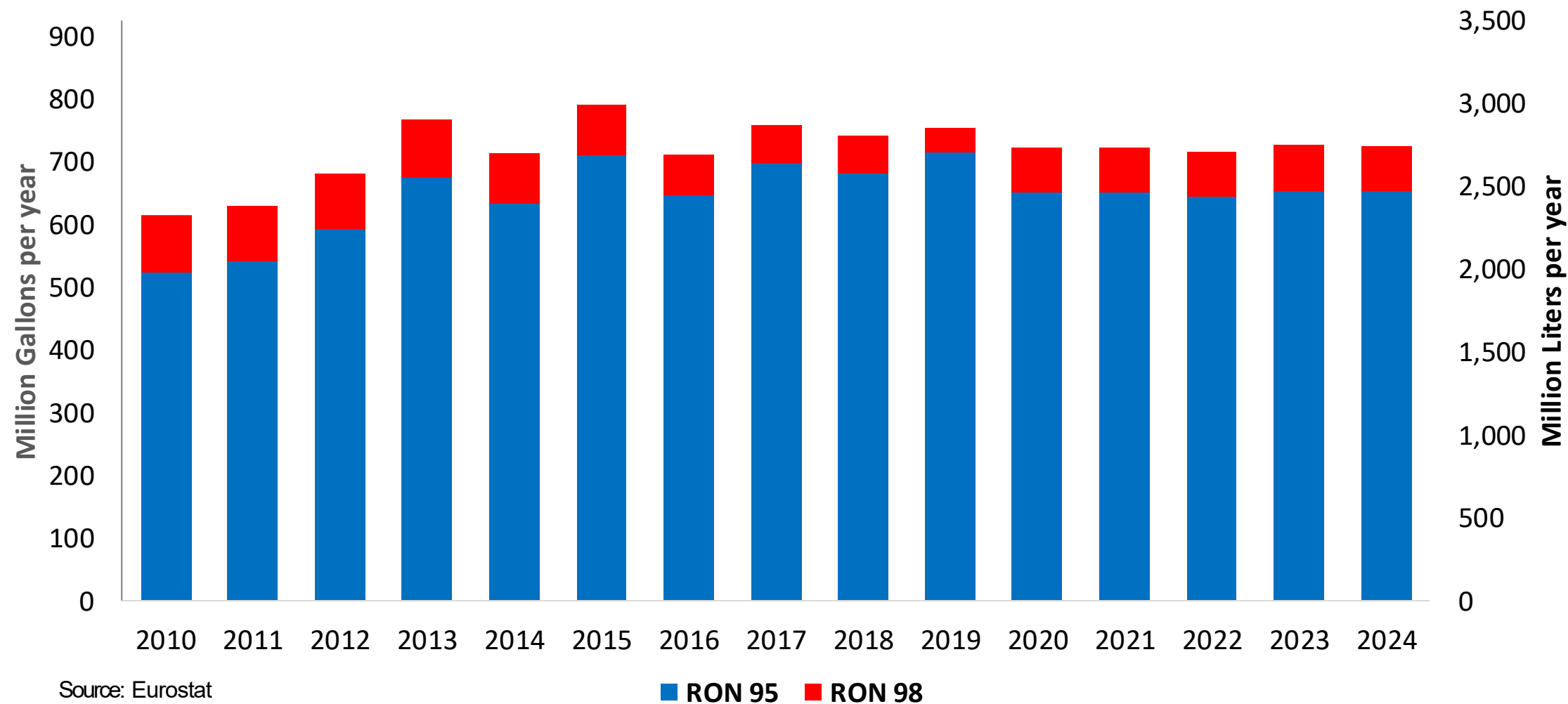
Gasoline Demand in Denmark



Gasoline Demand by Grade in Denmark

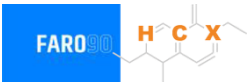


Gasoline Production by Grade in Denmark

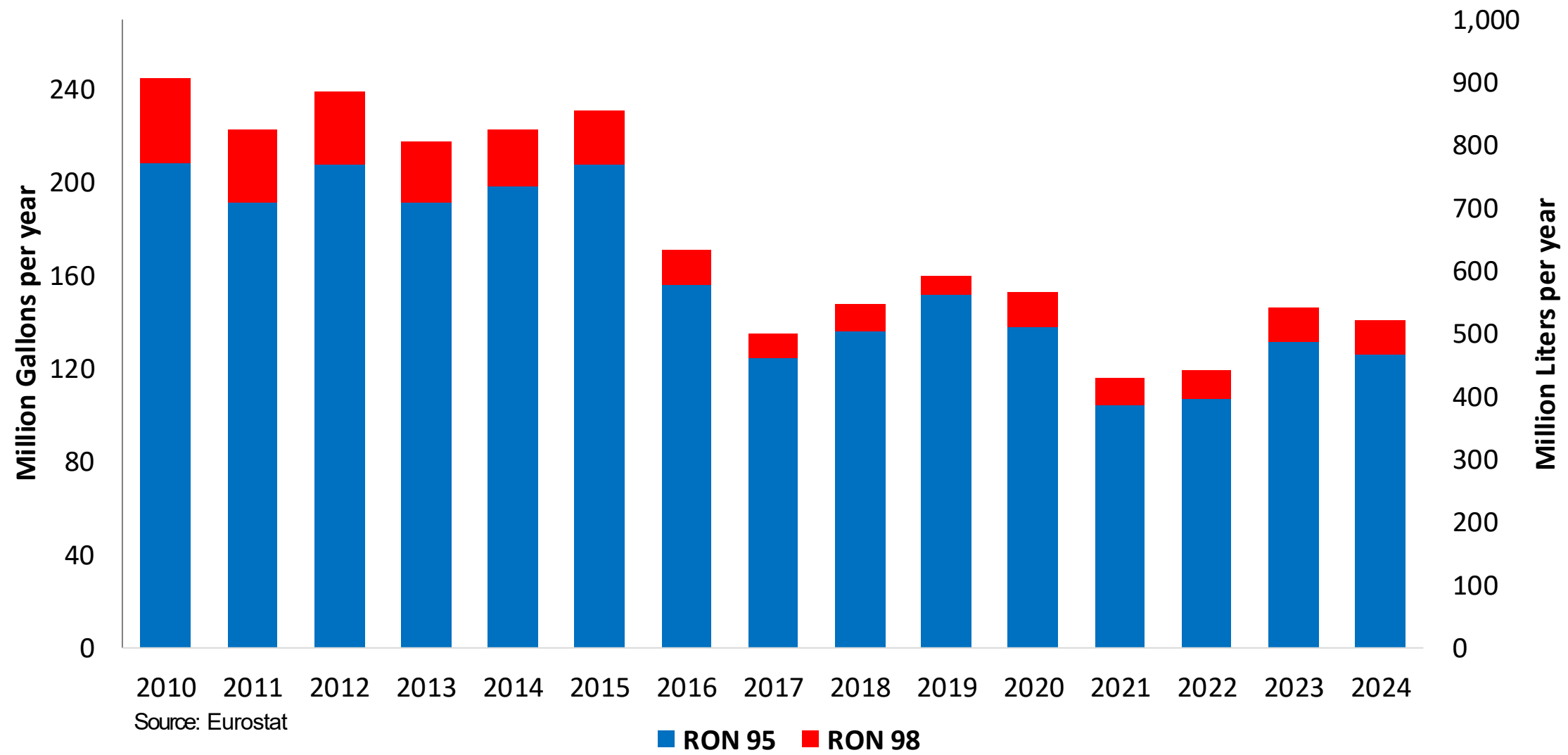


Source: Eurostat

Gasoline Imports in Denmark

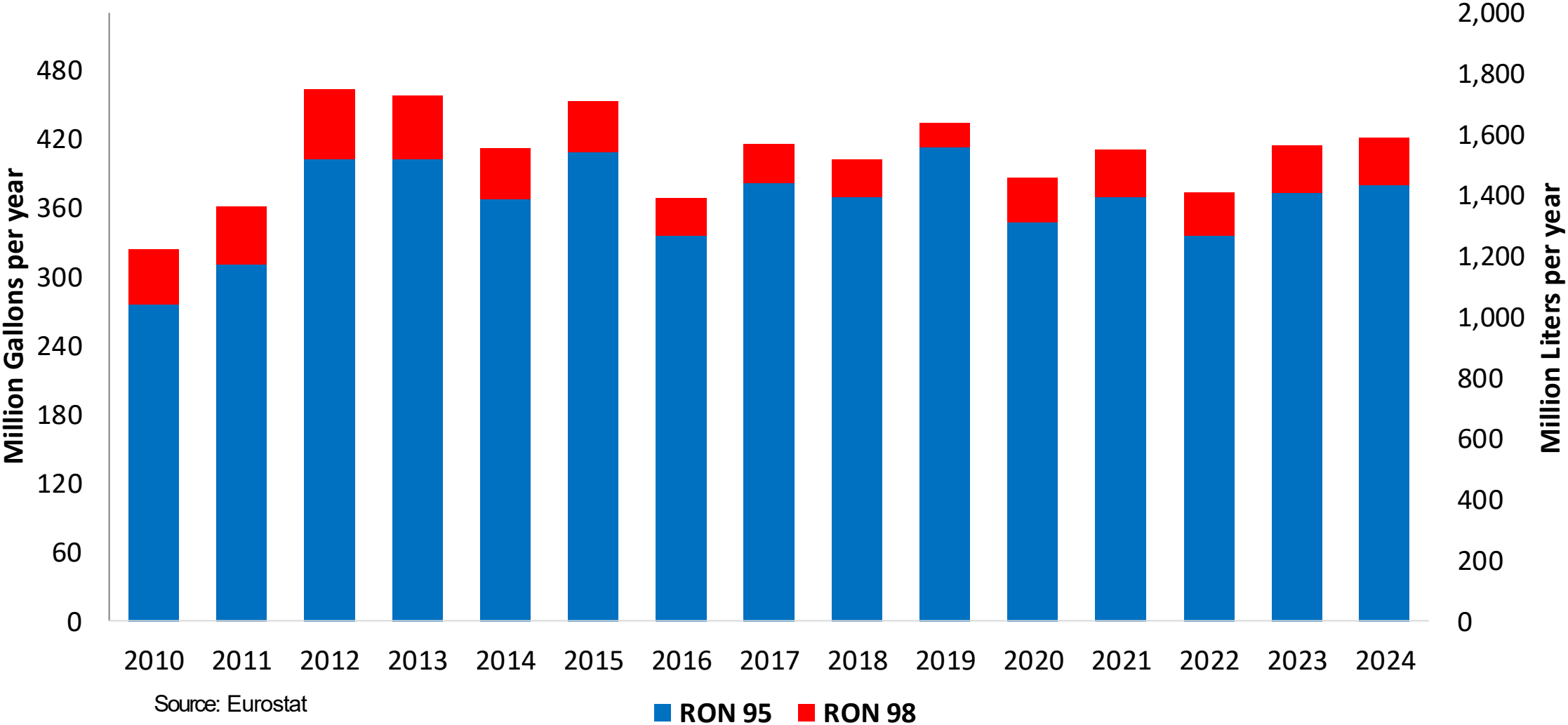


Gasoline Imports by Grade in Denmark

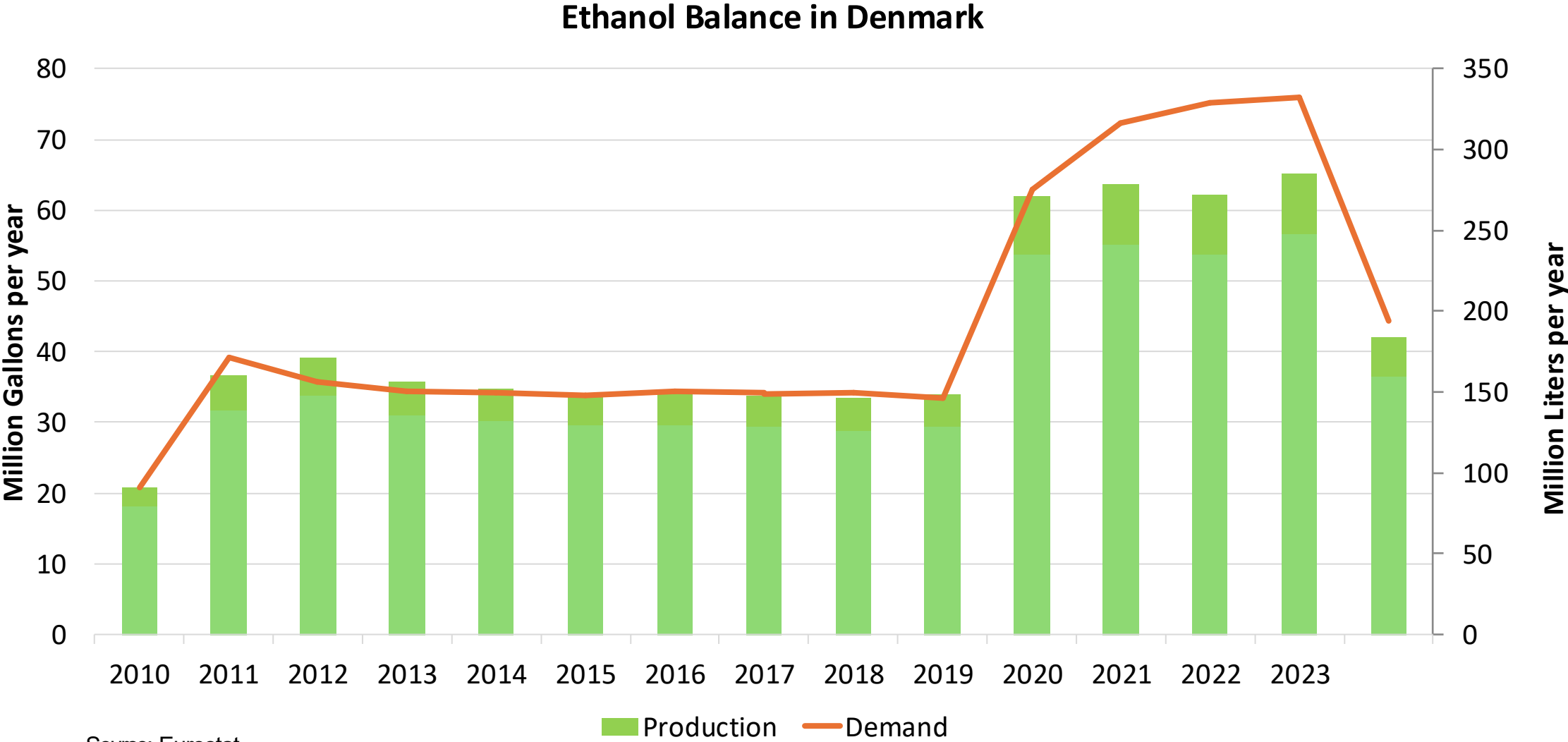


Gasoline Exports in Denmark

Gasoline Exports by Grade in Denmark



Ethanol Balance in Denmark



Source: Eurostat

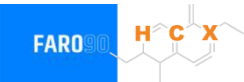
Gasoline Quality in Denmark

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	5.2							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

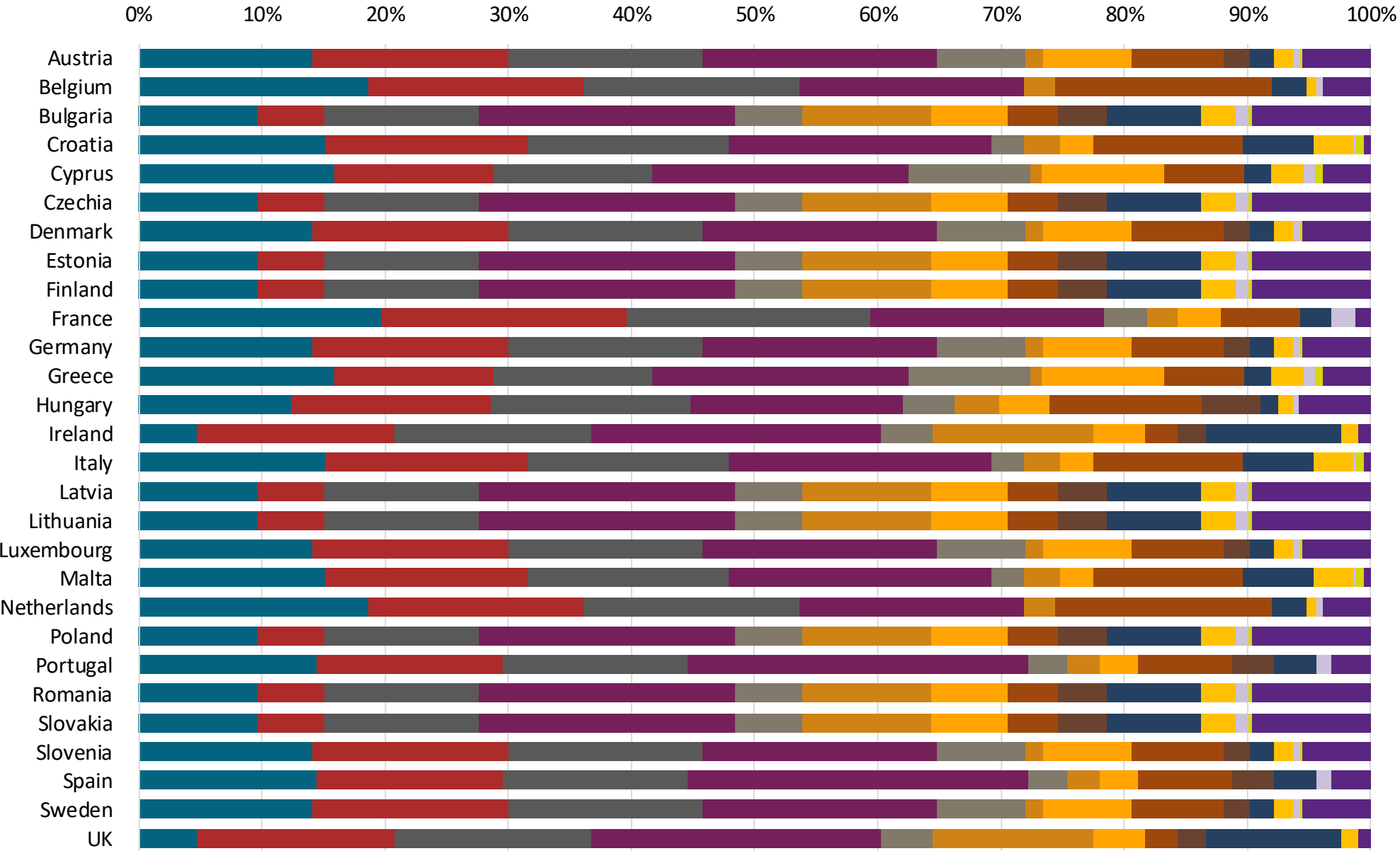
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source: Faro90



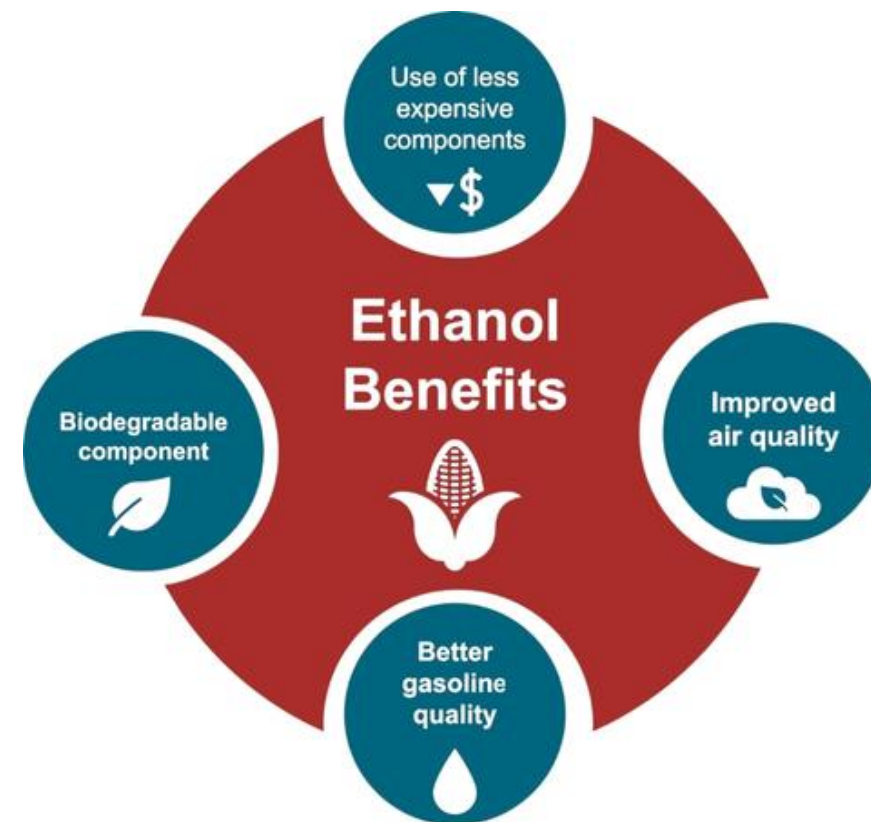
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

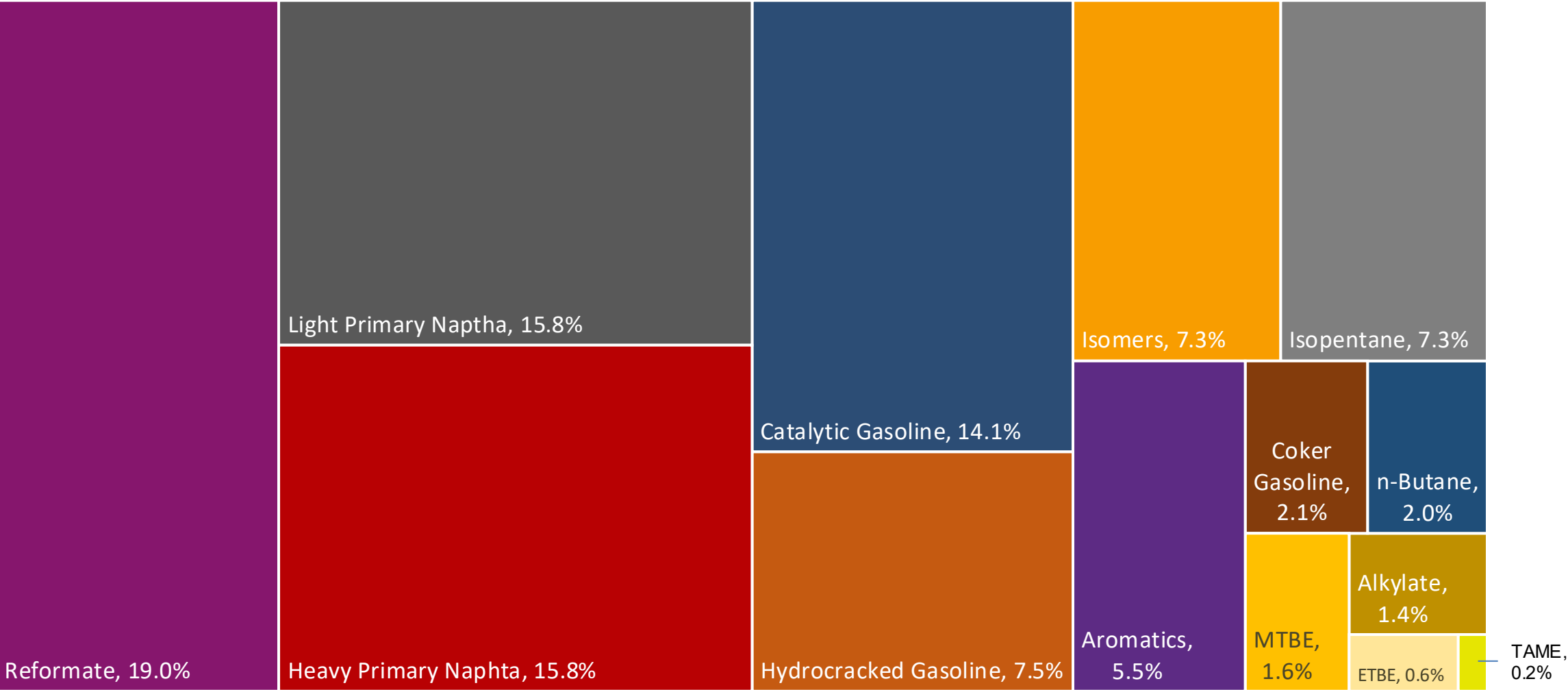
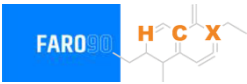
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

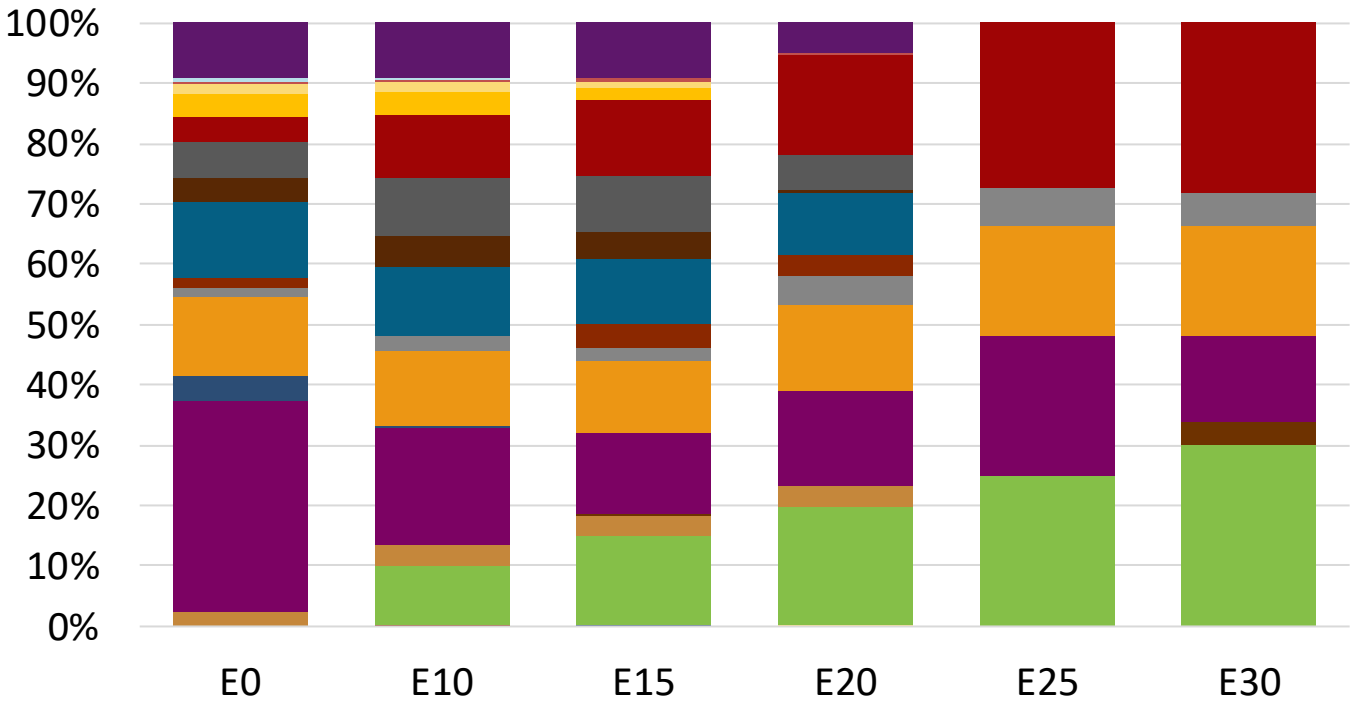
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Denmark Available Blending Components



Denmark – Gasoline 95 – Constant Octane

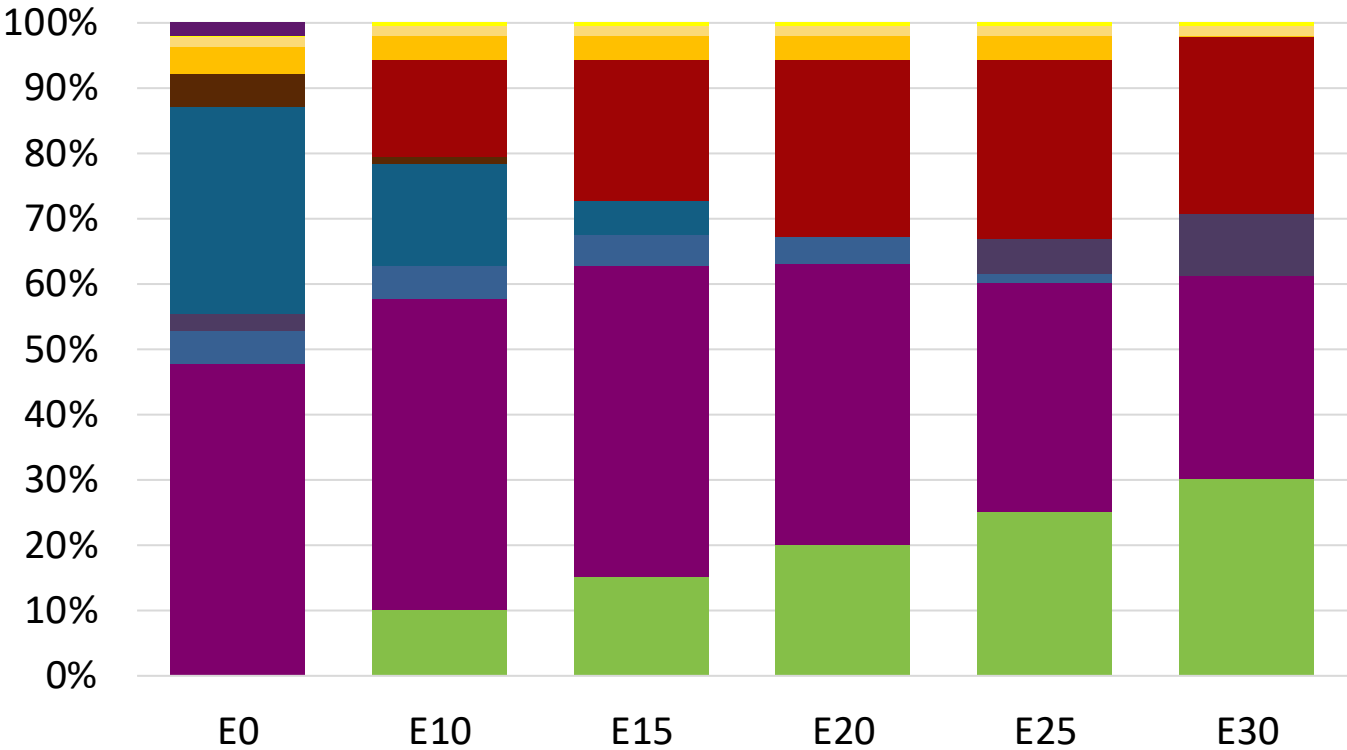
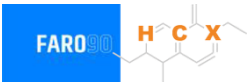


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US\$/gal)	\$ 2.272	\$ 2.144	\$ 2.083	\$ 2.002	\$ 1.880	\$ 1.814

Source: Faro90

Denmark - Gasoline 98 - Constant Octane

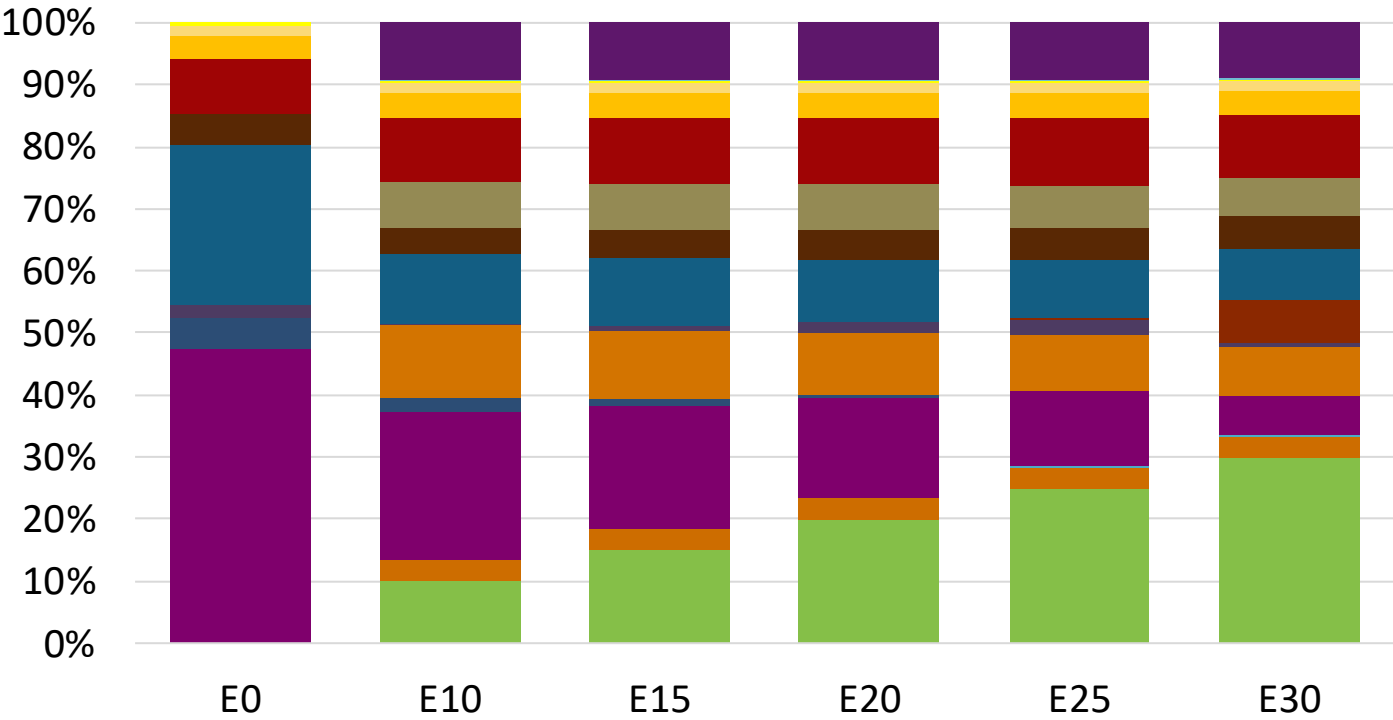


- Ethanol
- Alkylate
- Raffinate
- Reformate
- n-Butane
- Isomerate
- Isopentane
- Hydrocracked Gasoline
- Catalytic Gasoline
- Coker Naphtha
- Light Primary Naphtha
- Heavy Primary Naphtha
- MTBE
- ETBE
- TAME
- Aromatics

Average CIF 2024
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.347	\$ 2.218	\$ 2.146	\$ 2.069	\$ 1.992	\$ 1.925

Denmark – Gasoline 95 – Increased Octane

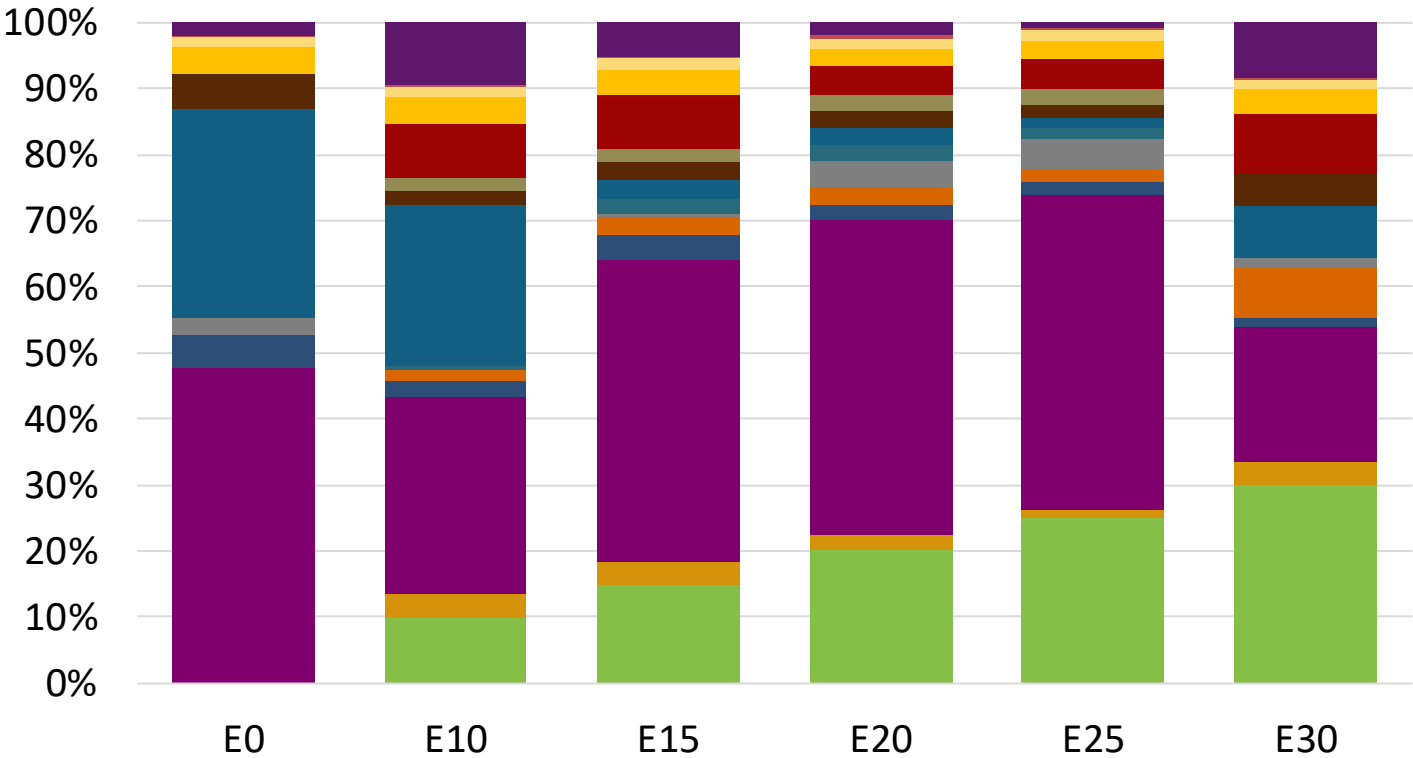
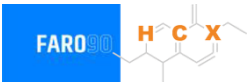


Average CIF 2024
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US\$/gal)	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209	\$ 2.209

Source: Faro90

Denmark – Gasoline 98 – Increased Octane



Average CIF 2024
Prices

Octane (RON)	98.0	99.2	101.3	102.6	104.0	104.7
Price (US\$/gal)	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

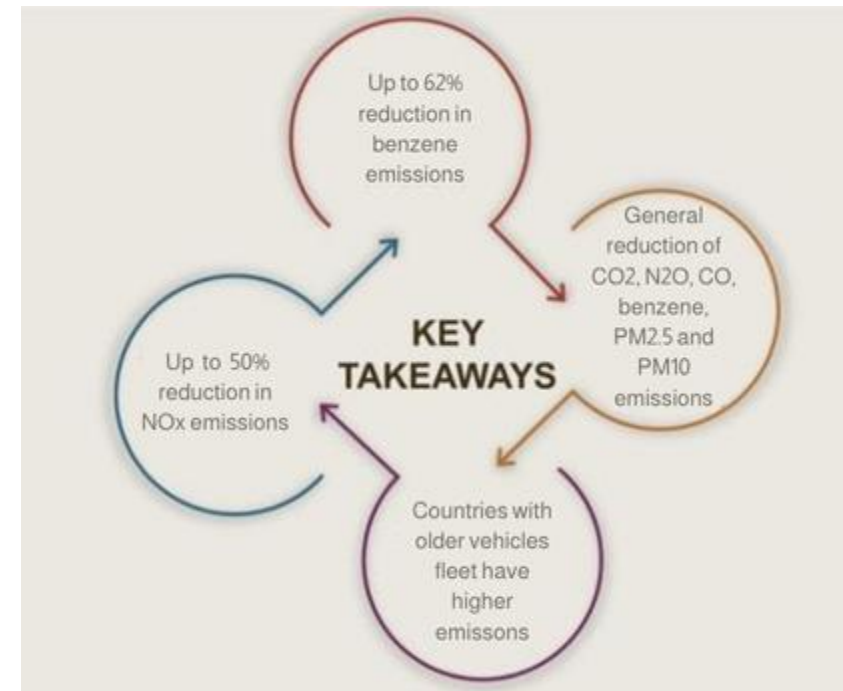
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

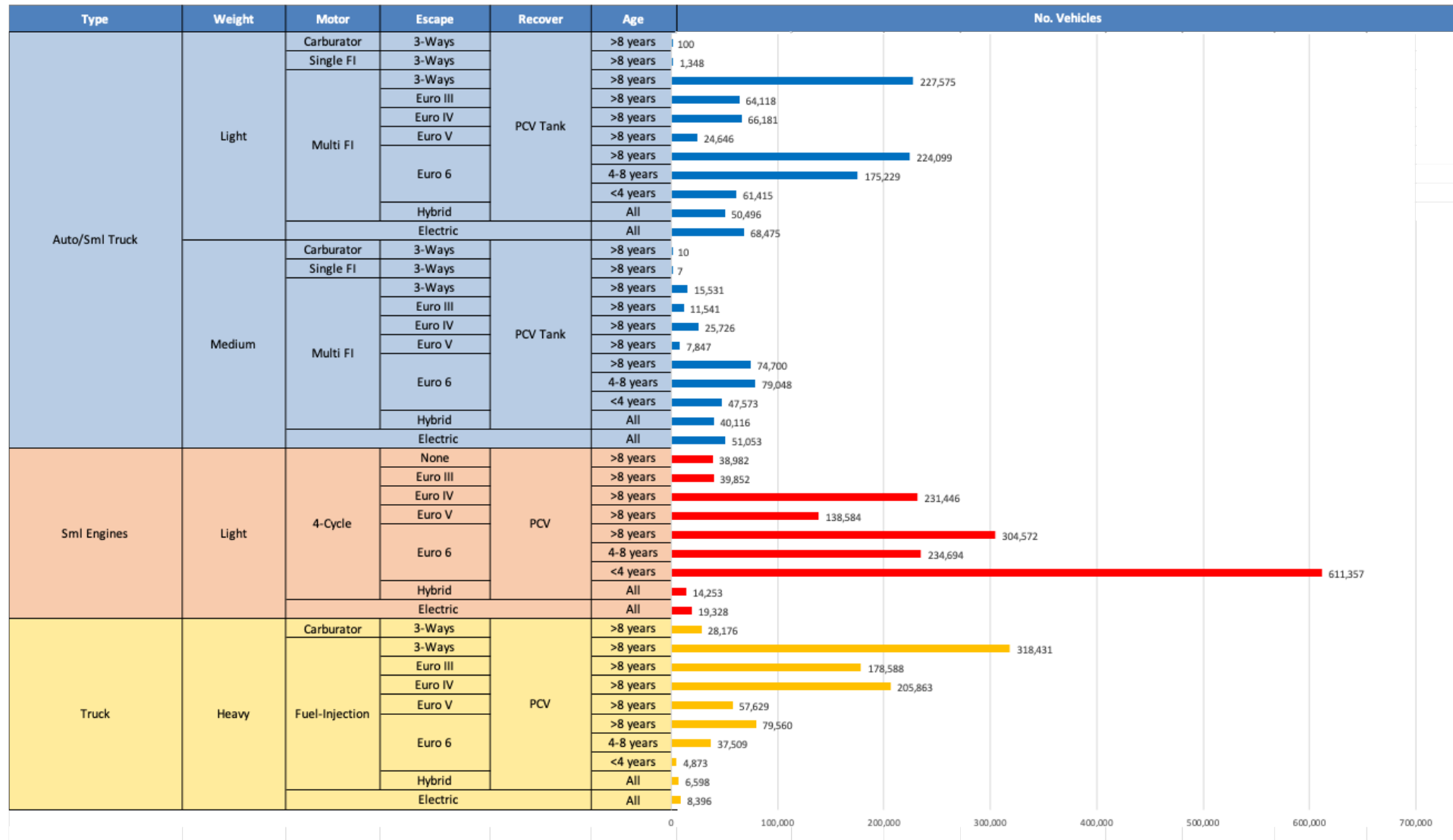
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



*Source: Bureau of transportation statistics.

Gasoline Vehicle Fleet - Denmark

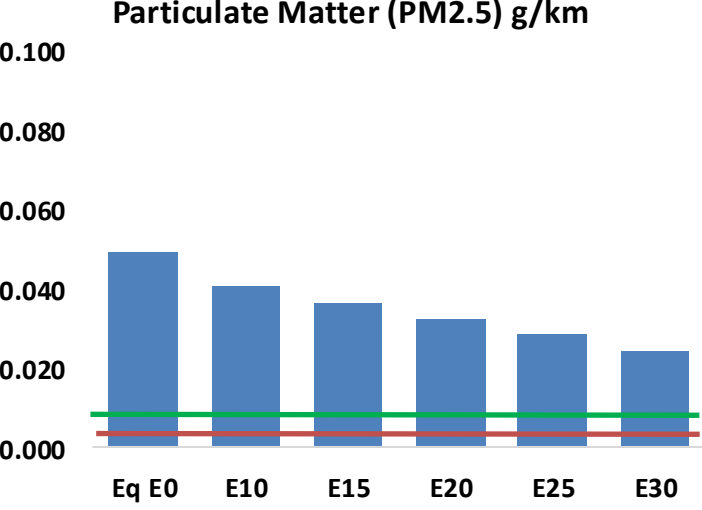
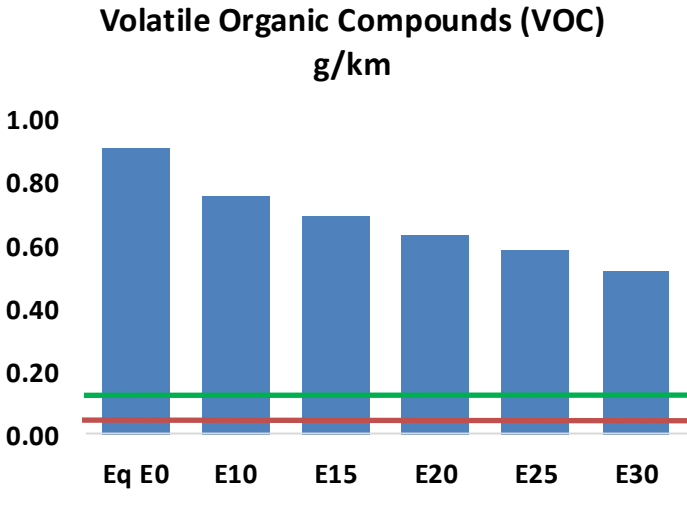
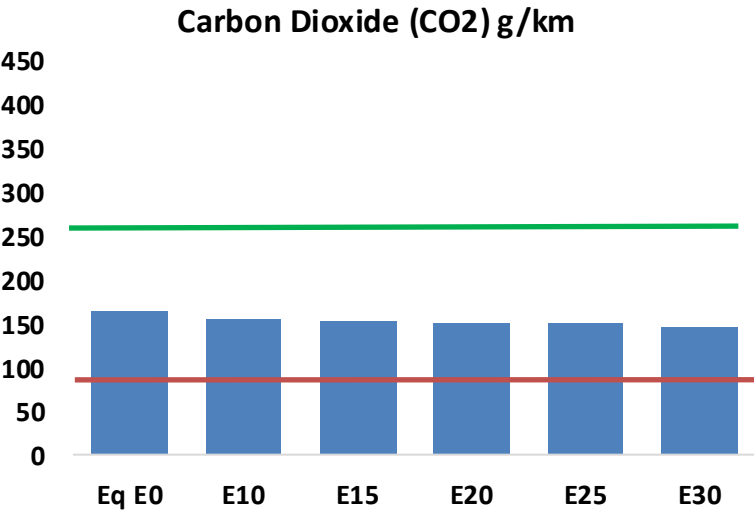
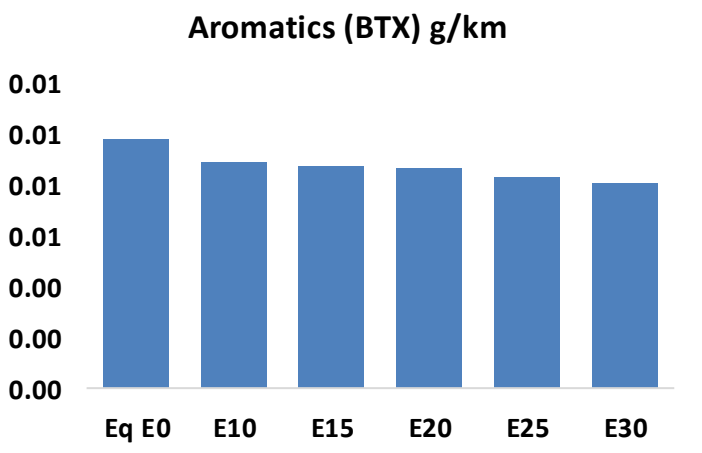
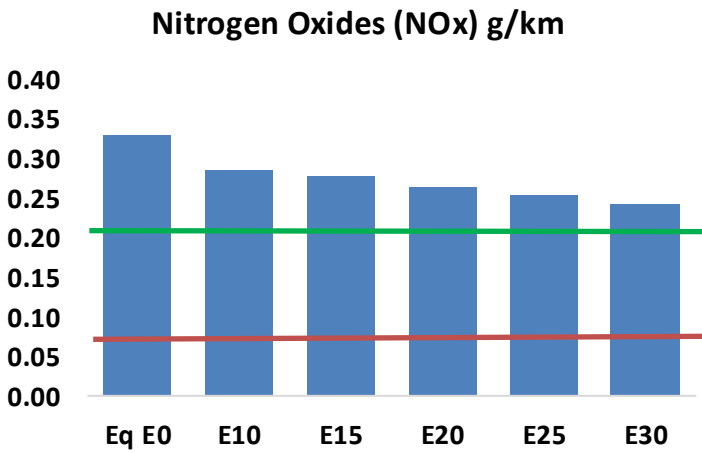
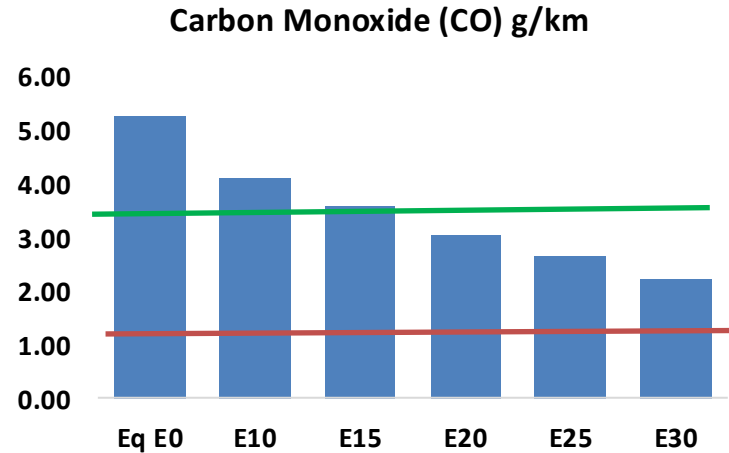
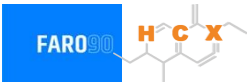


Vehicle Fleet: 3,875,526
Gasoline Vehicle Fleet: 2,055,566

Source: Eurostat, ACEA

Denmark – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Denmark – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	267,891	238,105	208,320	182,118	154,926	133,767	112,248	-22%	-42%
CO2	8,251,751	8,039,563	7,839,163	7,681,555	7,603,564	7,562,067	7,422,676	-5%	-8%
VOC	46,011	42,194	38,377	35,194	31,966	29,476	26,181	-17%	-31%
NOx	16,638	15,362	14,509	14,086	13,392	12,853	12,216	-13%	-20%
SOx	93	86	79	73	67	61	54	-15%	-28%
NH3	3,336	3,303	3,270	3,285	3,322	3,348	3,370	-2%	0%
N2O	140	139	138	139	142	143	145	-1%	2%
CH4	10,288	10,288	10,288	10,442	10,669	10,864	11,049	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	269	252	235	223	210	200	187	-12%	-22%
Acetaldehyde	842	966	1,418	2,159	2,942	3,362	3,972	68%	249%
Formaldehyde	3,272	3,181	3,709	4,355	4,562	5,047	5,494	13%	39%
Benzene	498	478	453	446	442	423	409	-9%	-11%
PM2.5	528	480	432	390	348	304	258	-18%	-34%
PM10	1,976	1,796	1,617	1,459	1,304	1,138	965	-18%	-34%

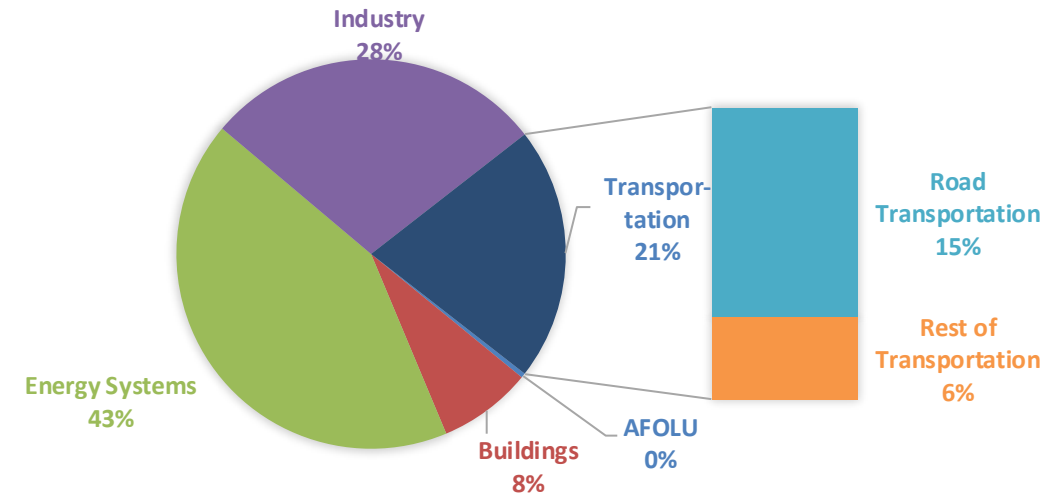
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

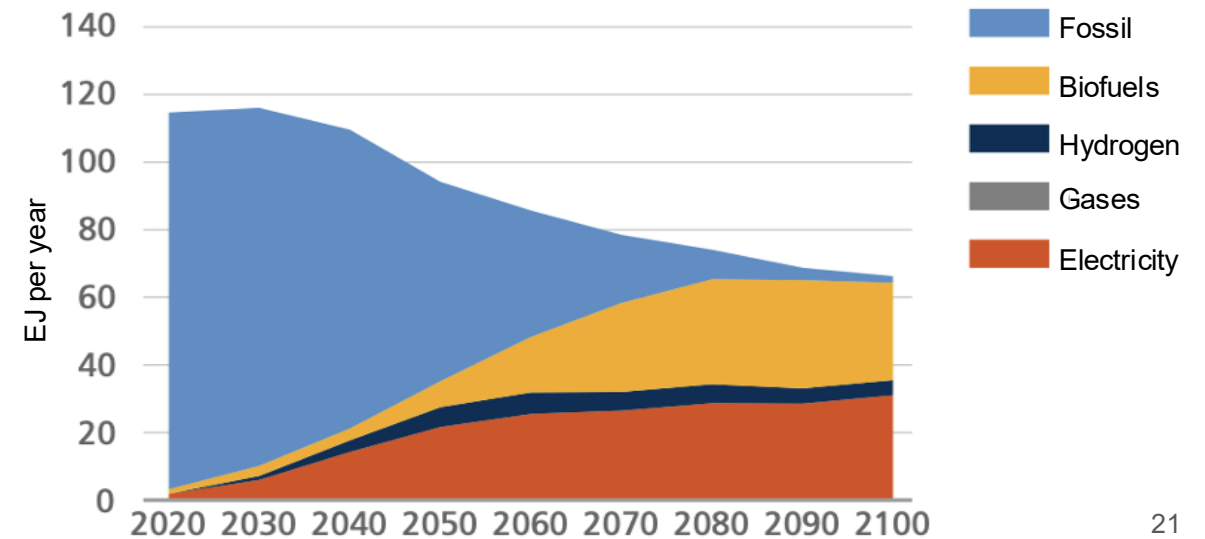
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

Current GHG Global Emissions Distribution

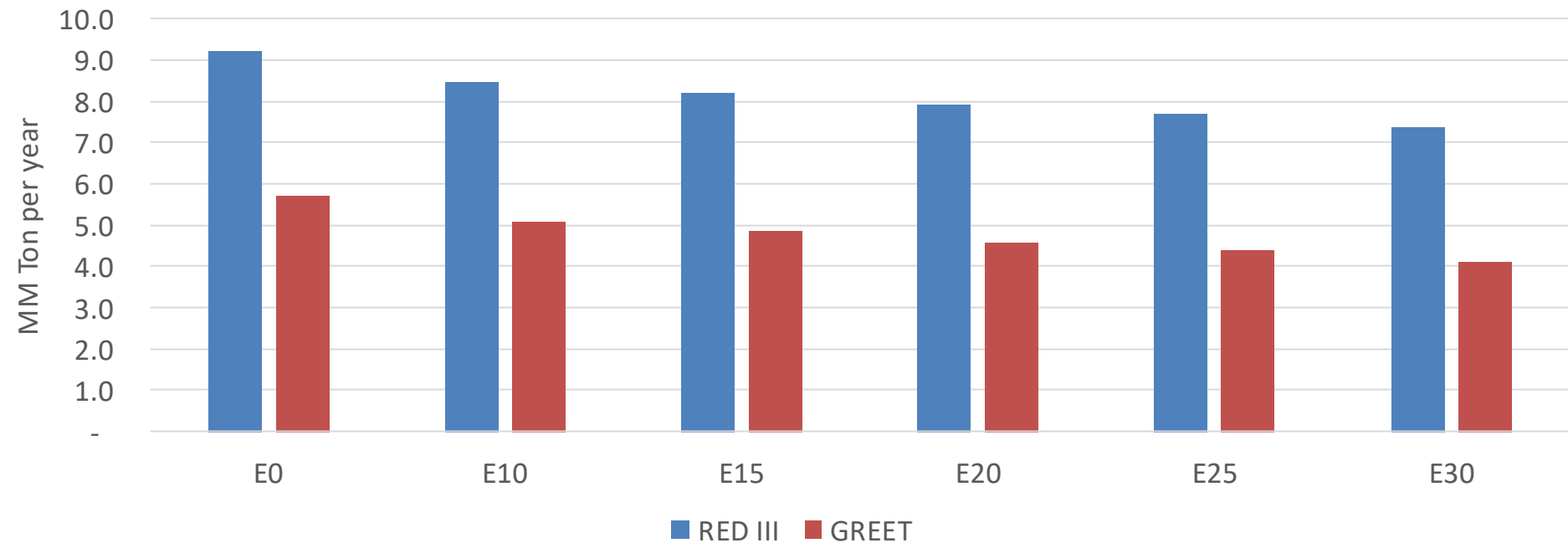


IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Denmark – Gasoline Vehicle Fleet Life Cycle GHG Emissions

Life Cycle GHG Emission - Gasoline Vehicles



Country Emissions 2024 (MMT/año)	45.5
Target @ 2035 (MMT/year)	25.6
Delta	19.9

vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.1%	15.2%	19.4%	23.2%	27.8%
RED II/III	0.0%	8.0%	11.0%	14.0%	16.7%	19.9%

Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.2%	4.4%	5.6%	6.6%	8.0%
RED II/III	0.0%	3.7%	5.1%	6.5%	7.7%	9.2%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90